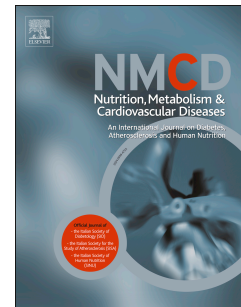


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Metabolic Abnormalities, But Not Obesity Per Se, Associated with Chronic Kidney Disease in a Taiwanese Population

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Acronyms: CKD, chronic kidney disease; DM, diabetes; HTN, hypertension; CAD,
coronary heart disease; CVA, cerebral vascular accident; BMI, body mass index;
MHNW, metabolically healthy normal weight; MHOW, metabolically healthy
overweight; MHO, metabolically healthy obese; MUNW, metabolically unhealthy
normal weight; MUOW, metabolically unhealthy overweight; MUO, metabolically
unhealthy obese.

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text is 3277. The number of references is 55. The number of the tables is 3.

Abstract**Background and Aims:**

It is inconclusive whether obesity itself or metabolic abnormalities are linked to chronic kidney disease (CKD). The aim of this study was to examine the association between different subtypes of obesity and metabolic abnormalities with CKD in adults.

Methods and Results:

This study enrolled 14983 eligible subjects stratified into metabolically healthy normal weight (MHNW), metabolically healthy overweight (MHOW), metabolically healthy obesity (MHO), metabolically unhealthy normal weight (MUNW), metabolically unhealthy overweight (MUOW), and metabolically unhealthy obesity (MUO) according to body mass index and metabolic syndrome status (ATP-III criteria).

The metabolic healthy phenotype was defined as the absence of both metabolic syndrome and any known diabetes, coronary artery disease, stroke, hypertension or dyslipidemia. Early and advanced CKD were defined as $eGFR < 60$, proteinuria, or structural abnormalities as detected by renal sonography.

The prevalence of CKD was 2.5, 3.0, 4.0, 10.6, 9.5, and 10.5% in subjects with MHNW, MHOW, MHO, MUNW, MUOW, and MUO, respectively. In the multivariate analysis, the MUNW (OR:2.22, $P < 0.001$), MUOW (OR:2.22, $P < 0.001$), and MUO (OR:2.45,

$P < 0.001$) groups were associated with early CKD. For advanced CKD, the OR was 2.56 ($P < 0.001$), 2.31 ($P < 0.001$), and 3.49 ($P < 0.001$) in the MUNW, MUOW, and MUO groups, respectively. The associated risks of early and advanced CKD were not significant in the MHOW and MHO group. MUOW and MUO were associated with higher risk of CKD compared with MHOW and MHO after adjusting other variables.

Conclusions:

Metabolic abnormalities, but neither overweight nor obesity, were associated with a higher risk of CKD in adults.

Introduction

Chronic kidney disease has become a major health issue because it leads to end-stage renal disease (ESRD) and is related to increased risk of cardiovascular diseases [1]. Metabolic syndrome (MetS), a grouping of various metabolic abnormalities, has been shown to be attributed to the development of CKD [2]. These metabolic abnormalities, including dyslipidemia, elevated blood pressure, and increased fasting blood sugar, are frequently found in obese individuals. Although obesity is associated with cardiovascular disease and has been reported to be a risk factor for CKD [3], there is still no final conclusion as to whether it is obesity itself or the metabolic abnormalities resulting from obesity that link to CKD.

Recent studies identified a subgroup of obese subjects with a low burden of metabolic abnormalities, as the so-called metabolically healthy obese (MHO) phenotype [4]. There have been some studies investigating the relationship between MHO and CKD, and their results were found to be inconsistent. One cohort study showed that overweight and obesity were associated with an increased incidence of CKD in metabolically healthy patients [5], and the other study reported a healthy metabolic profile does not protect obese adults from incident CKD [6]. In contrast, two other studies showed that the MHO phenotype was not associated with higher risk of CKD compared with metabolic healthy

non-obese individuals [7, 8] although one meta-analysis study reported that MHO individuals eventually have a higher risk for CKD [9]. Furthermore, most studies investigating the association between MHO and CKD only used $eGFR < 60$ [5, 6, 8, 10-13] as the definition of CKD; only a few studies included proteinuria [7, 14-16], and none used structural abnormalities detected by renal sonography. Therefore, we conducted a cross-sectional study to examine the association between MHO and CKD based on a more precise definition and a large sample size population.

Method

In this cross-sectional study, we enrolled a consecutive number of 16,537 subjects who underwent a health check-up at the Health Examination Center in National Cheng Kung University Hospital, Taiwan, between January 1, 2000 and August 31, 2009. We conducted a secondary data analysis without personal identification information, and informed consent was waived. The Institutional Review Board of National Cheng Kung University Hospital approved the study (approval number: B-ER-106-066). We excluded subjects aged < 18 years ($n = 23$) with body mass index (BMI) < 18.5 kg/m^2 ($n = 547$), heavy drinking (defined as more than 210g of alcohol per week, $n = 62$), renal transplantation ($n = 3$), pyuria ($n = 573$), bacteriuria ($n = 457$), and incomplete data regarding metabolic status ($n = 40$). A total of 14,983 subjects were included in the final analysis.

Anthropometric and Blood Pressure Measurements

All participants completed a standard structural questionnaire that included basic demographic data, medical history, medication history, alcohol consumption, smoking, and exercise habits. Smoking status and alcohol consumption status were both categorized as never, former, or current. Current smoking was at least 20 cigarettes per month for at least the previous 6 months. Current alcohol consumption was defined as at least once per week for at least the previous 6 months. Regular exercise was defined as a minimum of 20 minutes of regular exercise at least three times per week. Each subject's body height (to the nearest 0.1 cm), weight (to the nearest 0.1 kg), and waist circumference (to the nearest 0.1 cm) were measured. The waist circumference measurement was performed at the end of normal expiration on bare skin midway between the lower rib margin and the iliac crest along the mid-axillary line. The BMI was calculated as weight in kilograms divided by height in meters squared. The subject, who was in a fasting state, rested for at least 5 minutes in a supine position, and blood pressure was taken between 08:00 and 10:00 AM using a DINAMAP vital sign monitor (model 1846SX; Critikon Inc., Irvine, CA) .

Biochemistry and Sonography Measurements

After 10-h of overnight fasting, all participants underwent blood sampling for basic biochemical examinations, including fasting plasma glucose (FPG), glycated hemoglobin (HbA1c), creatinine, aspartate aminotransferase (AST), alanine aminotransferase (ALT), total cholesterol, triglyceride (TG), high-density lipoprotein cholesterol (HDL-C), and C-reactive protein (CRP). A first-voided, early-morning spot urine sample was obtained to measure urine protein (mg/dL). All subjects received an abdominal echo examination performed by experienced physicians using static grayscale and real-time B-mode units with a 3.5-MHz transducer after overnight fasting. Two-hour post-load glucose (2h-PG) was obtained using a 75-g oral glucose tolerance test in subjects without a known history of diabetes (DM).

Definition of CKD

The estimated glomerular filtration rate (eGFR) was derived from the 4-variable Modification of Diet in Renal Disease Study equation ($186 \times \text{Cr}^{-1.154} \times \text{age}^{-0.203} \times (0.742, \text{ if female}) \times (1.212, \text{ if African American})$) [17]. Proteinuria was defined as 30mg/dL or higher of protein in the morning urine using a dipstick test [18]. Structural abnormalities was detected using kidney sonography, and the following changes, including polycystic kidney, dysplastic kidney, hydronephrosis, and parenchymal disease were identified [18]. CKD was defined as $\text{eGFR} < 60 \text{ ml/min per } 1.73 \text{ m}^2$, proteinuria, or structural abnormalities. Early CKD was defined as CKD stage

1 (kidney damage with $\text{eGFR} \geq 90 \text{ ml/min/1.73 m}^2$) and 2 (kidney damage with eGFR of $60\text{--}89 \text{ ml/min/1.73 m}^2$), and advanced CKD was defined as CKD stage 3 to 5 (eGFR , $<60 \text{ ml/min/1.73 m}^2$) [18].

Definition of body weight state and metabolic phenotype

Body weight state was categorized according to the Taiwan BMI cutoffs: normal weight ($\text{BMI } 18.5\text{--}23.9 \text{ kg/m}^2$), overweight ($\text{BMI } 24\text{--}26.9 \text{ kg/m}^2$), and obesity ($\text{BMI} \geq 27 \text{ kg/m}^2$) [19]. MetS was defined according to criteria established by the National Cholesterol Education Program Adult Treatment Panel III [20], which requires the presence of three or more of the following components: 1) abdominal obesity by waist circumference (defined using Asian-specific waist circumference cutoff values of $\geq 90 \text{ cm}$ for men and $\geq 80 \text{ cm}$ for women), 2) high blood pressure (systolic blood pressure ≥ 130 or diastolic blood pressure $\geq 85 \text{ mmHg}$, 3) hypertriglyceridaemia (fasting plasma triglycerides $\geq 150 \text{ mg/dl}$), 4) low HDL cholesterol (fasting HDL cholesterol $< 40 \text{ mg/dl}$ in men, $< 50 \text{ mg/dl}$ in women), and 5) hyperglycemia (fasting plasma glucose $\geq 100 \text{ mg/dl}$). Metabolic healthy phenotype was defined as the absence of MetS and the following diseases: coronary artery disease (defined by positive medical/medication history), cerebrovascular accident (defined by positive medical/medication history), dyslipidemia (defined by positive medical/medication history), hypertension (defined by $\text{SBP} \geq 140 \text{ mmHg}$, $\text{DBP} \geq 90 \text{ mmHg}$ or positive

medical/medication history), and DM (defined by FPG $\geq$ 126 mg/dl, 2h-PG $\geq$ 200 mg/dl, A1c $\geq$ 6.5%, or positive medical/medication history). Based on the BMI status and metabolic phenotypes, all study subjects were classified into 6 groups: metabolically healthy normal weight (MHNW), metabolically healthy overweight (MHOW), metabolically healthy obese (MHO), metabolically unhealthy normal weight (MUNW), metabolically unhealthy overweight (MUOW), and metabolically unhealthy obesity (MUO).

Statistics

All continuous variables were expressed as means $\pm$ standard deviations (SD). All categorical variables were presented as numbers and proportions. The analyses of the continuous and categorical variables used to assess differences among the six groups, including MHNW, MHOW, MHO, MUNW, MUOW, and MUO were examined using a one-way ANOVA or chi-square test, respectively. Post hoc tests were used for multiple comparisons of continuous variables among groups.

The multivariate-adjusted logistic regression models were used to investigate the associations between different body weights and metabolic statuses with CKD. Model 1 was adjusted for age and sex, and Model 2 was further adjusted for history of CAD (coronary heart disease), CVA (cerebral vascular accident), alcohol consumption, smoking, and regular exercise, in addition to the adjusted variables in

Model 1. Model 3 was adjusted for the variables in Model 2, plus ALT, and the presence of the highest quartile of CRP (≥ 2.2 mg/L). All statistical analyses were performed using SPSS software (version 17.0; SPSS, Chicago, IL). A *P*-value less than 0.05 was considered statistically significant.

Results

Overall, 14,983 participants were included. The mean age was 49.4 ± 12.5 years, and the mean BMI was 24.6 ± 3.4 kg/m². The prevalence of metabolic syndrome, overweight, and obesity were 24.3% (3648/14983), 33.0% (4948/14983), and 21.4% (3200/14983), respectively. There were 889 (5.9%) participants diagnosed with CKD based on eGFR, proteinuria, and image abnormality. The prevalence of CKD was 2.5% in the MHNW group, 3.0% in the MHOW group, 4.0% in the MHO group, 10.6% in the MUNW group, 9.5% in the MUOW group, and 10.5% in the MUO group ($p < 0.001$, test for trend). Furthermore, 1.7%, 1.7%, 2.7%, 5.0%, 5.2%, and 5.7% of the participants in the MHNW, MHOW, MHO, MUNW, MUOW, and MUO groups were found to have early CKD, respectively. The prevalence of advanced CKD was 0.8%, 1.3%, 1.3%, 5.6%, 4.4%, and 4.8% in the MHNW, MHOW, MHO, MUNW, MUOW, and MUO groups, respectively.

The characteristics of all the participants are shown in Table 1. The proportion of MHNW, MHOW, MHO, MUNW, MUOW, and MUO subjects was 33.9 %, 17.1 %, 17.1 %, 17.1 %, 17.1 %, and 17.1 %, respectively.

6.9 %, 11.7 %, 16.0%, and 14.5%, respectively. The following variables were significantly different among subjects with different body weights and metabolic phenotypes (Table 1): gender, age, body weight, BMI, waist circumference, systolic blood pressure, diastolic blood pressure, total cholesterol, TG, HDL-C, FPG, HbA1c, 2h-PG, creatinine, eGFR, AST, ALT, highest CRP quartile, regular exercise, tobacco use, alcohol consumption, and prevalence of CKD, HTN, DM, dyslipidemia, CAD, and CVA. Based on the post-hoc analysis, compared with MUO individuals, MHO subjects were characterized by younger age, lower BMI, and a more favorable risk profile (lower waist circumference, blood pressure, HbA1c, 2h-PG, fasting glucose, total cholesterol, and higher HDL-C). In addition, there was an increased trend of serum creatinine level and a decreased trend of eGFR among groups from MHNW to MUO.

Initially, the multiple logistic regression analysis revealed that the metabolically unhealthy phenotypes were all associated with higher risk of CKD. The associated risks in the fully adjusted model were 2.28 (95% CI 1.78-2.92, $P<0.001$), 2.12 (95% CI 1.67-2.69, $P<0.001$), and 2.63 (95% CI 2.08-3.34, $P<0.001$) in the MUNW, MUOW, and MUO groups, respectively (Table 2). When CKD was further categorized as early and advanced stages, the metabolically unhealthy subgroups were still associated with a higher prevalence of CKD regardless of the

stage of CKD when compared with MHNW (Table 3). The associated risk of early CKD in the fully adjusted model was 2.22 (95% CI 1.62-3.06, $P < 0.001$), 2.22 (95% CI 1.65-2.99, $P < 0.001$), and 2.45 (95% CI 1.82-3.29, $P < 0.001$) in the MUNW, MUOW, and MUO groups, respectively. For advanced CKD, the odds ratio was 2.56 (95% CI 1.73-3.77, $P < 0.001$) in the MUNW group, 2.31 (95% CI 1.57-3.40, $P < 0.001$) in the MUOW group, and 3.49 (95% CI 2.37-5.15, $P < 0.001$) in the MUO group. In contrast, the risk of early and advanced CKD was not significant in the MHOW and MHO groups, as compared to the MHNW group. In addition, MUO (early CKD: OR 1.91, 95% CI 1.26-2.92, $P = 0.003$; advanced CKD: OR 2.75, 95% CI 1.52-5.00, $P = 0.001$) was found to be associated with higher risk of CKD compared with MHO after adjusting other variable (data not shown). Similarly, MUOW (early CKD: OR 2.73, 95% CI 1.91-3.91, $P < 0.001$; advanced CKD: OR 1.76, 95% CI 1.16-2.65, $P < 0.001$) was also positively associated with CKD when compared with MHOW (data not shown). Thus, given the same category of BMI, metabolically unhealthy subjects had a higher associated risk of CKD than their counterparts. In the fully adjusted model, age, male gender, and highest quartile of CRP were significantly associated with increased risk of early CKD. Age, male gender, and highest quartile of CRP were positively associated with advanced CKD. ALT and alcohol consumption were negatively related to advanced CKD.

Discussion

Although previous studies have investigated the association between obesity and CKD, the role of obesity in CKD is still unclear. Some studies reported that obesity is linked to an increased risk of CKD [3, 21-24], but some did not find this association after adjustment for cardiovascular disease risk factors [25, 26]. Up to the present, prior studies examining the association between MHO and CKD have also yielded contradictory results [5-8, 10, 12, 14, 16]. This study found that subjects with MHO and MHOW were not associated with a higher risk of CKD compared to healthy normal weight subjects. On the other hand, subjects with the metabolic unhealthy phenotype (MUNW, MUOW, and MUO) all were found to have an increased risk of early and advanced CKD. One possible explanation for the inconsistent results is related to different definitions of CKD because many studies did not adopt proteinuria and structural changes in the kidney as indicators. None of the aforementioned studies applied structural changes in the kidney when defining CKD, and this might have resulting in their missing the early stage of CKD. Another explanation may be related to the different metabolic status of the obesity phenotypes and the lack of consensus for the definition of MHO. As for the definition of MHO, most studies [5, 6, 8, 12, 14, 16, 27-29] applied Adult Treatment Panel III (NECP-ATP-III) criteria, while one [7] applied IDF criteria, and one [10] used

other criteria. For those studies adopting ATPIII criteria, three [8, 12, 16] studies used ≥ 3 of the 5 metabolic components, three [6, 27, 29] excluded waist circumference and applied ≥ 2 of the 4 metabolic components, one [28] included hs-CRP, and two [5, 14] included insulin resistance. In addition to MetS, cardiovascular diseases and metabolic disorders, such as hypertension, diabetes and dyslipidemia, are required to be considered when defining MHO, but only one [8] study excluded diabetes from MHO, and one [5] excluded hypertension, diabetes and dyslipidemia. We think excluding these metabolic disorders from MHO is crucial, because in the sensitivity analysis, we found that MHO, defined as Nam et al's study [29], were also positively related to CKD. Based on a relatively large, healthy population without DM, HTN, dyslipidemia, CAD and CVA, this study provided a more accurate estimation of the relationship between MHO and CKD. After fully considering the potential for bias, it was determined that the detrimental effect of metabolic abnormalities seems to have more influence than obesity per se on CKD.

The association between obesity and CKD might be mediated through multiple biologic mechanisms, including hormonal factors, inflammation, and possible vascular endothelial growth factor [30]. As for hormonal factors, the literature has supported the role of insulin resistance and related abnormalities on kidney

dysfunction [31, 32]. MHO individuals exhibit less insulin-resistance than their MUO counterparts [33]. Furthermore, leptin has direct effects on renal pathophysiology [34, 35]. Leptin was significantly higher in the MUO individuals than in the MHO individuals. In addition, an inverse correlation between adiponectin and proteinuria were found [36], and adiponectin was higher in individuals with a metabolically healthy phenotype than in the metabolic abnormal individuals regardless of the degree of obesity [37]. Additionally, it was proposed that chronic inflammation may initiate kidney injury in patients with obesity. eGFR was found to be inversely associated with circulating inflammation markers [38]. In addition, there was a distinct up-regulation of inflammatory cytokines and their related molecules in the glomeruli of patients with obesity-related glomerulopathy [30]. The MHO phenotype, with lower concentrations of CRP, IL-6, and TNF- α , has been shown to exhibit a more favorable inflammatory status than the MUO phenotype [39, 40]. VEGF may contribute to endothelial dysfunction and may also play an important role in the development of pathological changes in patients with obesity-related glomerulopathy, especially the formation of glomerular hypertrophy [30]; however, the role of VEGF in the various metabolic phenotype needs more elucidation [41]. In summary, the elevated insulin resistance, leptin, inflammation markers, and possible VEGF, but a low level of adiponectin may partially explain why the MUNW, MUOW,

and MUO groups were associated with increased risk of CKD, but the MHNW, MHOW and MHO groups were not in this study. Further studies are needed to determine the mechanism underlying the link among obesity, metabolic abnormalities, and CKD.

In this study, age, male sex and high CRP level were associated with increased risk of CKD in both early and advanced diseases. Age is a known independent risk factor for renal impairment [42]. The protective effects of estrogen in women and/or the damaging effects of testosterone, together with unhealthier lifestyles in men, might cause kidney function to decline faster in men than in women [43]. This study also showed that increased CRP was associated with early and advanced CKD, which was consistent with the findings of prior studies indicating that systemic inflammation is a predictive factor for the development of CKD [44, 45]. Additionally, in this study, lower levels of ALT were found to be related to increased associated risk of advanced CKD. Previous study has revealed that serum ALT was low in CKD with and without ESRD as compared with a control group [46], and the level of ALT tended to decrease in proportion to CKD progression [47]. The association of alcohol consumption with CKD has been conflicting [48, 49]. After excluding subjects with heavy drinking, we found alcohol use to be associated with a decreased risk of advanced CKD. An inverse association of alcohol consumption and CKD has been

found, but this association was only present in those who drank moderately [50]. In our study, exercise was associated with a lower risk of CKD, and the result was consistent with the findings of a previous study [51, 52]. However, when CKD was divided by stage, exercise was insignificantly related to both early and advanced CKD, which may have been related to inadequate case numbers of early and advanced CKD patients. Smoking is a well-known risk factor of CKD [53, 54], but smoking was not significantly associated with CKD in our study. This inconsistent finding may be related to the fact that smokers tend to quit smoking when they are informed of the disease, resulting in an insignificant association between smoking and CKD.

There are some limitations in this study. First, it was a cross-sectional study, so we could not confirm the cause-and-effect relationship between MHO and CKD. Second, we used single measurement of eGFR and proteinuria to diagnose CKD, and this may lead to misclassification of CKD. Third, we could not assess the change of the obese-metabolic phenotypes through time. For example, MHNW might be MHOW or MHO previously, and MHO was deemed as a dynamic process by some studies [4, 55], so further study will be needed to investigate the influence of such changes to the obese-metabolic phenotypes during the life on CKD. Fourth, sonography is a skill-dependent procedure, so there might be inter-physicians difference when making the diagnosis of structural abnormalities. Last, the study

subjects were confined to Taiwanese participants receiving a health check-up, and obesity and MetS were defined according to Taiwan and Asia-Pacific criteria, respectively, so it's uncertain whether the findings can be generalized to other ethnic groups.

In conclusion, the MHO phenotype was not associated with an increased risk of CKD, but metabolically abnormal phenotypes were. The detrimental effect of metabolic abnormalities seems to be more obvious than obesity per se on CKD. Longitudinal studies are needed as to the association between the changes of obese-metabolic phenotypes and CKD.

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Conflict of interest:

The authors declare no conflict of interest.

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Author contribution

H.C., J.W., and Y.C. were involved in the conception and design of the study and interpretation of the data. J.W., Y.Y., F.L. and C.C. provided administrative, technical, and material support. R.W. provided technical and material support. All authors were involved in the acquisition of data. H.C., J.W., and Y.C. undertook the data analysis. Drafting of the manuscript was done by H.C., J.W.

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Table 1. Baseline characteristics of participants according to body weight and metabolic phenotype

Variables	Metabolically healthy			Metabolically unhealthy			P value
	MHNW (n=5075)	MHOW (n=2558)	MHO (n=1033)	MUNW (n=1760)	MUOW (n=2390)	MUO (n=2167)	
Male	2397(47.2)	1876(73.3)	701(67.9)	1031(58.6)	1751(73.3)	1601(73.9)	<0.001
Age (year)	44.5(11.5)	47.0(11.0)	46.4(12.0)	57.2(11.8)	55.4(11.3)	52.1(11.8)	<0.001
BW (kg)	58.2(7.5)	69.5(7.2)	80.2(10.6)	58.6(7.3)	69.3(7.4)	80.9(10.7)	<0.001
BMI (kg/m ²)	21.7(1.5)	25.3(0.8)	29.2(2.3)	22.2(1.4)	25.5(0.8)	29.6(2.5)	<0.001
WC (cm)	76.1(6.6)	85.5(5.7)	93.7(8.1)	80.4(6.7)	88.7(5.8)	96.9(7.8)	<0.001
SBP (mmHg)	109.4(11.6)	114.3(10.9)	115.9(10.6)	130.4(21.2)	129.2(19.0)	131.0(18.0)	<0.001
DBP (mmHg)	64.8(8.4)	68.3(8.2)	68.6(8.3)	75.3(11.6)	76.3(11.3)	77.1(11.2)	<0.001
TC (mg/dL)	190.6(35.7)	197.1(35.3)	194.0(33.5)	201.7(40.0)	204.9(39.2)	204.1(37.4)	<0.001
HDL-C (mg/dL)	54.4(14.1)	47.6(11.2)	46.3(10.2)	48.9(14.4)	43.4(11.7)	40.5(10.0)	<0.001
TG (mg/dL)	98.7(53.1)	121.4(60.7)	120.7(62.6)	141.1(106.0)	174.9(128.3)	193.3(115.6)	<0.001
FPG (mg/dL)	85.9(7.9)	88.4(7.9)	88.8(7.5)	106.8(42.1)	106.2(35.5)	107.9(33.4)	<0.001
2h-PG (mg/dL)	104.4(28.6)	111.2(29.6)	113.8(30.9)	149.8(77.2)	151.7(67.1)	156.4(67.8)	<0.001
HbA1c (%)	5.4(0.3)	5.5(0.3)	5.6(0.3)	6.3(1.6)	6.2(1.3)	6.2(1.2)	<0.001
AST (U/L)	23.2(11.5)	26.1(17.0)	27.5(13.3)	28.1(30.7)	29.7(21.9)	32.1(18.7)	<0.001
ALT (U/L)	24.4(19.1)	33.9(39.5)	38.0(29.6)	32.2(45.8)	39.3(42.0)	47.3(35.3)	<0.001
Highest quartile of CRP	1010(19.9)	829(32.4)	469(45.4)	615(34.9)	982(41.2)	1211(55.9)	<0.001
Regular exercise	376(7.4)	227(8.9)	74(7.2)	170(9.7)	211(8.8)	181(8.4)	0.020
Smoking status							

Never	4360(85.9)	2077(81.2)	859(83.2)	1414(80.3)	1866(78.1)	1661(76.6)	<0.001
Quitted	248(4.9)	194(7.6)	61(5.9)	149(8.5)	245(10.3)	211(9.7)	<0.001
Current	467(9.2)	286(11.2)	113(10.9)	197(11.2)	278(11.6)	294(13.6)	<0.001
Alcohol intake							
Never	4492(88.5)	2182(85.3)	874(84.6)	1480(84.1)	1924(80.5)	1750(80.8)	<0.001
Quitted	132(2.6)	80(3.1)	33(3.2)	94(5.3)	111(4.6)	98(4.5)	<0.001
Current	447(8.8)	294(11.5)	125(12.1)	185(10.5)	352(14.7)	315(14.5)	<0.001
SCr (mg/ dL)	0.83(0.23)	0.92(0.25)	0.89(0.21)	0.94(0.88)	0.94(0.40)	0.95(0.37)	<0.001
eGFR (ml/min per 1.73 m ²)	96.3(17.5)	91.4(16.3)	92.9(17.3)	88.7(21.1)	87.1(17.7)	88.2(18.8)	<0.001
Proteinuria	76(1.5)	30(1.2)	22(2.1)	89(5.1)	118(4.9)	125(5.8)	<0.001
Structural abnormalities	32(0.6)	20(0.8)	11(1.1)	57(3.2)	59(2.5)	44(2.0)	<0.001
CKD	129(2.5)	77(3.0)	41(4.0)	187(10.6)	228(9.5)	227(10.5)	<0.001
Stage I ~II	88(1.7)	43(1.7)	28(2.7)	88(5.0)	124(5.2)	124(5.7)	<0.001
Stage III~V	41(0.8)	34(1.3)	13(1.3)	99(5.6)	104(4.4)	103(4.8)	<0.001
HTN	0(0)	0(0)	0(0)	896(50.9)	1156(48.4)	967(44.6)	<0.001
Dyslipidemia	0(0)	0(0)	0(0)	381(21.6)	478(20.0)	366(16.9)	<0.001
DM	0(0)	0(0)	0(0)	627(35.6)	776(32.5)	726(33.5)	<0.001
CAD	0(0)	0(0)	0(0)	135(7.7)	149(6.2)	94(4.3)	<0.001
CVA	0(0)	0(0)	0(0)	31(1.8)	34(1.4)	12(0.6)	<0.001

Data are expressed as means (SD) or number (percentage). Abbreviations: BW, body weight; BMI, body mass index; WC, waist circumference;

SBP, systolic blood pressure; DBP, diastolic blood pressure; TC, total cholesterol; HDL-C, high-density lipoprotein cholesterol; TG, triglyceride;

LDL-C, low-density lipoprotein cholesterol; FPG, fasting plasma glucose; 2h-PG, two-hour post-load glucose; HbA1c, hemoglobin A1c; AST, aspartate aminotransferase; ALT, alanine aminotransferase; Scr, serum creatinine; eGFR, estimated glomerular filtration rate; CKD, chronic kidney disease; HTN, hypertension; DM, diabetes; CAD, coronary heart disease; CVA, cerebral vascular accident; MHNW, metabolically healthy normal weight; MHOW, metabolically healthy overweight; MHO, metabolically healthy obese; MUNW, metabolically unhealthy normal weight; MUOW, metabolically unhealthy overweight; MUO, metabolically unhealthy obese.

Table 2. Adjusted odds ratios and 95% confidence interval for the risk of CKD according to body weight and metabolic phenotype

Variable	MHNW	MHOW	MHO	MUNW	MUOW	MUO
Model 1	1.00(ref.)	1.02 (0.76-1.36)	1.38 (0.96-1.99)	2.37* (1.86-3.03)	2.27* (1.80-2.86)	2.99* (2.38-3.77)
Model 2	1.00(ref.)	1.02 (0.76-1.36)	1.38 (0.96-1.99)	2.41* (1.88-3.07)	2.30* (1.82-2.91)	3.03* (2.41-3.82)
Model 3	1.00(ref.)	0.95 (0.71-1.27)	1.26 (0.87-1.81)	2.28* (1.78-2.92)	2.12* (1.67-2.69)	2.63* (2.08-3.34)

Data expressed as OR (95% CI), * $p < 0.001$

Abbreviations: CKD, chronic kidney disease; MHNW, metabolically healthy normal weight; MHOW, metabolically healthy overweight; MHO, metabolically healthy obese; MUNW, metabolically unhealthy normal weight; MUOW, metabolically unhealthy overweight; MUO, metabolically unhealthy obese; ref., reference.

Model 1 was adjusted for age and sex.

Model 2 was adjusted for the variables in Model 1, plus alcohol consumption, smoking status and exercise.

Model 3 was adjusted for the variables in Model 2, plus alanine aminotransferase (ALT), and highest quartile of C-reactive protein (≥ 2.2 mg/L)

Table 3. Adjusted odds ratios and 95% confidence interval for the risk of early and advanced CKD according to body weight and metabolic phenotype

Variable	MHNW	MHOW	MHO	MUNW	MUOW	MUO
Model 1						
Stage I~II	1.00(ref.)	0.88 (0.61-1.27)	1.46 (0.95-2.25)	2.37* (1.73-3.26)	2.45* (1.83-3.29)	2.93* (2.20-3.91)
Stage III~V	1.00(ref.)	1.38 (0.87-2.20)	1.29 (0.68-2.44)	2.54* (1.73-3.74)	2.28* (1.56-3.34)	3.43* (2.36-5.00)
Model 2						
Stage I~II	1.00(ref.)	0.88 (0.61-1.27)	1.46 (0.95-2.25)	2.40* (1.75-3.30)	2.49* (1.86-3.34)	2.95* (2.21-3.94)
Stage III~V	1.00(ref.)	1.39 (0.87-2.21)	1.30 (0.68-2.45)	2.58* (1.75-3.80)	2.31* (1.58-3.39)	3.50* (2.41-5.10)
Model 3						
Stage I~II	1.00(ref.)	0.81 (0.56-1.18)	1.28 (0.83-1.98)	2.22* (1.62-3.06)	2.22* (1.65-2.99)	2.45* (1.82-3.29)
Stage III~V	1.00(ref.)	1.32 (0.82-2.11)	1.27 (0.67-2.41)	2.56* (1.73-3.77)	2.31* (1.57-3.40)	3.49* (2.37-5.15)

Data expressed as OR (95% CI), * $p < 0.001$

Abbreviations: CKD, chronic kidney disease; MHNW, metabolically healthy normal weight; MHOW, metabolically healthy overweight; MHO,

metabolically healthy obese; MUNW, metabolically unhealthy normal weight; MUOW, metabolically unhealthy overweight; MUO, metabolically unhealthy obese; ref., reference.

Model 1 was adjusted for age and sex.

Model 2 was adjusted for the variables in Model 1, plus alcohol consumption, smoking status and exercise.

Model 3 was adjusted for the variables in Model 2, plus alanine aminotransferase (ALT), and highest quartile of C-reactive protein (≥ 2.2 mg/L)

Highlights

- In metabolically healthy subjects, obesity and overweight were not associated with an increased risk of CKD compared normal weight.
- Metabolically unhealthy phenotypes all were found to have an increased risk of early and advanced CKD compared to metabolically healthy normal weight.
- Given the same category of BMI, metabolically unhealthy subjects had a higher associated risk of CKD than their counterparts.